# PROGRAMMING IN JAVA

# CSA0915

# ASSIGNMENT-CO1

# CAMPUS SMART-PARKING SYSTEM

# TASK 07 – Method Overriding

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## 1. Task07.md

### Aim

To demonstrate method overriding in Java using the Campus Smart-Parking System.

### Method Overriding

Method overriding occurs when a child class provides its own implementation of a method already defined in its parent class.

### Rules

1. The method name must be the same.
2. The method parameters must be the same.
3. Inheritance is required.
4. The `@Override` annotation is used to indicate overriding.

### Application

The Vehicle class contains a parking fee method. The Car and TwoWheeler classes override it to calculate different parking fees.

### Advantages

- Provides specialized behaviour.
- Supports runtime polymorphism.
- Improves code reusability.

### Conclusion

Method overriding allows different vehicle types to calculate parking fees using their own implementations.

## 2. Vehicle.java

```
public class Vehicle {
    void calculateFee() {
        System.out.println("General Parking Fee: Rs. 30");
    }
}
```

## 3. Car.java

```
public class Car extends Vehicle {
   @Override
    void calculateFee() {
        System.out.println("Car Parking Fee: Rs. 50");
    }
}
```

## 4. TwoWheeler.java

```
public class TwoWheeler extends Vehicle {
    @Override
    void calculateFee() {
        System.out.println("Two-Wheeler Parking Fee: Rs. 20");
    }
}
```

## 5. Main.java

```
public class Main {
    public static void main(String[] args) {
        Vehicle v1 = new Vehicle();
        Vehicle v2 = new Car();
        Vehicle v3 = new TwoWheeler();
        System.out.println("Campus Parking Fee Details:");
       v1.calculateFee();
        v2.calculateFee();
        v3.calculateFee();
    }
}
```

## 6. Output:

```
Campus Parking Fee Details:

General Parking Fee: Rs. 30

Car Parking Fee: Rs. 50

Two-Wheeler Parking Fee: Rs. 20
```